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On the hierarchy of plate models for a singularly perturbed multi-well nonlinear elastic energy

Edoardo Giovanni Tolotti[†]

Abstract

In the celebrated work of Friesecke, James and Müller '06 the authors derive a hierarchy of models for plates by carefully analyzing the Γ-convergence of the rescaled nonlinear elastic energy. The key ingredient of their proofs is the rigidity estimate proved in an earlier work of theirs. Here we consider the case in which the elastic energy has a multi-well structure: this type of functional arises, for example, in the study of solid-solid phase transitions. Since the rigidity estimate fails in the case of compatible wells, we follow Alicandro, Dal Maso, Lazzaroni and Palombaro '18 and add a regularization term to the energy that penalizes jumps from one well to another, leading to good compactness properties. In this setting we recover the full hierarchy of plate models with an explicit dependence on the wells. Finally, we study the convergence of energy minimizers with suitable external forces and full Neumann boundary conditions. To do so, we adapt the definition of optimal rotations introduced by Maor, Mora '21.

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Introduction

A large variety of elastic models for thin plates, i.e., slender bodies whose reference configuration has no intrinsic curvature, have been proposed in the mechanical literature (see for example [9, 2, 20, 25]). The objective of this work is to rigorously derive some of these models as Γ-limits of three-dimensional nonlinear elasticity. We focus on elastic energy densities that are minimized on a finite number of copies of SO(3). This setting is relevant, for example, in the modeling of solid-solid phase transitions (see [4]).

To be more precise, we consider a thin cylinder $\Omega_h = S \times (-\frac{h}{2}, \frac{h}{2})$, where the mid-plane S is a subset of $\mathbb{R}^2$ and h is the small thickness of the plate. Its elastic energy takes the form

$$E_h(w_h) = \int_{\Omega_h} W(\nabla w_h)\, dx\,,$$

where $w_h : \Omega_h \to \mathbb{R}^3$ is a deformation of Ω_h and W is the elastic energy density. We suppose that W is minimized on the set $K = \cup_{i=1}^{l} \mathrm{SO}(3)U_i$, where $U_1, \dots, U_l$ are symmetric and positive-definite

[†]Dipartimento di Matematica, Università di Pavia, Italy.
Email address: edoardogiovann.tolotti01@universitadipavia.it

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matrices. In this work, we study the Γ-limit of $\frac{1}{h^{\alpha+1}}E_h$, as $h \to 0$, for $\alpha \geq 2$. Roughly speaking, this corresponds to the analysis of deformations whose energy per unit volume $V_h = h^{-1}E_h$ scales like h^α.

This problem has been extensively studied in the mathematical literature. The regime $\alpha = 0$, i.e., $V_h \sim 1$, leads to the so-called membrane theory, rigorously derived for the first time by Le Dret and Raoult in [19] (see also [14, 3]). Here the hypotheses on W are compatible with a multi-well structure. The scaling $0 < \alpha < 2$ is largely unexplored and only partial results are available (see [10]). In the single-well setting (that is, $l = 1$ and $U_1 = \mathrm{Id}$), the cases $V_h \sim h^\alpha$, where $\alpha \geq 2$, have been studied by Friesecke, James, and Müller in [16, 15]. When $\alpha = 2$, they retrieve the so-called Kirchhoff-Love theory, first proposed by Kirchhoff in [18]. When $\alpha = 4$, they obtain the Von Kármán model introduced in [25], whereas in the case $2 < \alpha < 4$ and $\alpha > 4$ they derive the so-called constrained and linearized Von Kármán model, respectively. The key ingredient of all the latter results is the celebrated rigidity estimate proved by the same authors in [15]. Indeed, a quantitative understanding of the closeness of deformations gradients to the set of rotations is needed in order to rigorously linearize the energy density near the identity.

Various extensions of the rigidity estimate are available in the multi-well case. For two strongly incompatible wells, the rigidity estimate has been proved by Chaudhuri and Müller in [6] (and later with a different proof by De Lellis and Székelyhidi in [12]). In this setting the same authors also investigated the scaling $\alpha = 1$ in [7]. A further generalization is given by Chermisi and Conti in [8], for a finite number of well-separated wells. However, it is well-known that the rigidity estimate fails without separation assumptions on the wells. Indeed, if two of the wells are rank-one connected, it is easy to construct continuous deformations with zero elastic energy, whose gradient oscillates between the two wells. In the absence of a rigidity estimate, energy bounds fail to provide the necessary compactness for deformations.

One possible way to overcome this issue is to singularly perturb the elastic energy by adding a higher order term of the form

$$h^{-1}\eta^p(h)\int_{\Omega_h}|\nabla^2 w_h|^p\,dx\,,$$

where $\eta(h) \to 0$ as $h \to 0$ and $p > 1$ is a suitable exponent. This is a classical way of selecting preferred configurations in the Van der Waals–Cahn–Hilliard theory. Indeed, this additional term introduces a competition in the minimization problem: the free energy favors deformations with gradient in K, while the perturbation penalizes transitions between the wells.

Various analyses have been carried out in this setting. In the membrane scaling $\alpha = 0$, a full description of the Γ-limit of the perturbed energy with $p = 2$ is obtained in [13, 11]. As in the non-perturbed case, the hypotheses on the energy density allow for W to have a multi-well structure. The expression of the Γ-limit depends on the behavior of $\eta(h)/h$. We also recall the work by Shu [23], where an additional homogenization parameter is taken into account.

In our work we focus on the regimes $\alpha \geq 2$, and we do not assume any hypotheses of well-separation or connectedness of the wells. We follow ideas from Alicandro, Dal Maso, Lazzaroni, and Palombaro in [1], where the linearization of a multi-well elastic energy is studied. We show that the perturbation coefficient $\eta(h)$ can be chosen in such a way that, at the limit, deformation gradients are forced to fall into a single well and at the same time the (rescaled) perturbation term becomes negligible. Under these scaling assumptions on $\eta(h)$, in Section 3 we show that the L^2-norm of the distance of the deformation gradient from a specific well can be bounded by a suitable power of the perturbed elastic energy. In particular, once such a well has been identified, the usual rigidity estimate can be applied to deduce compactness.

The hierarchy of plate models that we derive is similar to the one found in [16, 15] for the single-well case. A key difference with respect to these works is the dependence of the limit

models on the well around which we have to linearize. For $\alpha = 2$, we retrieve the Kirchhoff model (see Theorem 1.1). Sequences of deformations with $V_h \sim h^2$ converge to isometric embeddings of the mid-plane into $\mathbb{R}^3$, for a flat metric depending on the well. The resulting model is similar in spirit to the one obtained in [5], where the authors consider a prestrained model and the limit deformation realizes an isometric embedding of the mid-plane for a metric depending on the prestrain. For $2 < \alpha < 4$, the Γ-limit is given by the constrained Von Kármán model, where the Von Kármán equations now depend on the well and take the form

$$\nabla u^T + \nabla u + |U_i^{-1} e_3|^2 \nabla v \otimes \nabla v = 0\,. \tag{1}$$

Here u and v are the limiting in-plane and out-of-plane displacements with respect to a reference deformation depending on the well (i.e., of the form $x \mapsto U_i x$). Equation (1) means that u and v satisfy a matching isometry condition up to the second order, for a well-dependent flat metric (see Section 1.2.2). For $\alpha = 4$ and $\alpha > 4$, we retrieve the Von Kármán and the linearized Von Kármán model, respectively. All the Γ-convergence results for $\alpha > 2$ are stated in Theorem 1.3. The Von Kármán model we obtain is similar to the one derived in [22] in the prestrained case.

Concerning the proofs, once compactness is established, the liminf inequality can be obtained arguing as in [16, 15]. Since the penalty term requires additional regularity, the truncation argument used in [16, 15] for the construction of the recovery sequences cannot be applied and is replaced by suitable approximation results. In particular, for $2 \leq \alpha < 4$ we need to assume some higher regularity for the mid-plane S.

The last part of the paper is devoted to the convergence of (quasi-)minimizers for the pure traction problem. More precisely, we consider a sequence of dead loads f_h and add to the energy their work per unit volume given by

$$-h^{-1} \int_{\Omega_h} f_h \cdot w_h \, dx\,.$$

We assume f_h to be of order $h^{\frac{\alpha}{2}+1}$. For $\alpha > 2$ we expect the limiting force term to depend only on the out-of-plane displacement (which is of order $h^{\frac{\alpha}{2}-1}$) and not on the in-plane displacement (which is of order $h^{\alpha-2}$ for $2 \leq \alpha < 4$, $h^{\frac{\alpha}{2}}$ for $\alpha \geq 4$). However, since the plate is not clamped, we need to understand how the presence of the force term reduces the rotation invariance of the problem. To do so, we follow the approach by Maor and Mora [21], where the notion of optimal rotations is introduced. These are rotations preferred by the forces, that can be different from the rotations selected by the rigidity estimate, around which linearization takes place. We show how this concept can be adapted to the dimension reduction setting. In this framework, we deduce a minimization property for the limit of (quasi-)minimizing sequences. A precise statement of these results is given in Theorems 1.5 and 1.7.

The paper is organized as follows. In Section 1 we introduce some notations, and we state the main results. After proving some preliminary lemmas, in Section 3 we prove compactness of sequences with bounded perturbed energy. In Section 4 we conclude the proof of the Γ-convergence results. Finally, in Section 5 we study the minimizing sequences in presence of dead loads. The Appendix A collects some results on isometries construction.

1 Notation and main results

1.1 Notation and functional setting

Consider a set of invertible matrices $U_1, \ldots, U_l \in \mathbb{R}^{3\times 3}$ with positive determinant and associate to each of them an energy well $K_i = \mathrm{SO}(3)U_i$. We assume the wells to be disjoint, i.e., $U_j^{-1}U_i \notin \mathrm{SO}(3)$